

AI Interaction Log

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Task / Prompt

The backend tests (PR #4) wouldn't run locally. Diagnose why, get them passing, and pin the Node version so the team doesn't hit the same wall.

Context / Findings

- `npm test` failed on a fresh checkout: first `Cannot find package 'supertest'` (deps not installed), then, after `npm install`, `Could not locate the bindings file` and finally `slice is not valid mach-o file` for `better-sqlite3`'s native binding.
- Root cause was the Node version, not the PR. Node had been installed via `brew install node`, which gave Node 26. `better-sqlite3@11.10.0` has no prebuilt binary for Node 26 and its source fails to compile against Node 26's V8 (no member named `'GetPrototype'` / `'GetIsolate'` / `'This'`). The current library (13.0.3) also declares `engines: >=22`.
- A secondary issue: the project lives on an iCloud path containing spaces, which broke node-gyp's shell invocation during the source-compile fallback.
- An earlier AI attempt to "rebuild" the binding ran in a Linux helper VM and produced a Linux binary, which macOS then rejected — a reminder that native modules are per-platform and must be verified on the target OS.

AI Output / Actions

- Walked through removing the bad setup: `delete backend/node_modules`, `git restore package.json` (npm had injected an `allowScripts` block), `brew uninstall node`, and clear the npm / node-gyp caches.
- Recommended running the project on Node 22 LTS via nvm. Confirmed by Bryce: on Node 22 `npm install` uses the prebuilt binary (no compile, no `allowScripts` step needed) and all 7 tests pass.
- Pinned the runtime on the branch to prevent recurrence:
 - Added `backend/.nvmrc = 22`.
 - Added `"engines": { "node": ">=22 <25" }` to `backend/package.json`.
- Decided the `install-scripts approve` step does NOT belong in the README, since it was only a symptom of being on Node 26 and is unnecessary on Node 22.

Decision

Bryce: Followed the cleanup steps, moved back to Node 22 LTS via nvm, and confirmed the install and all seven tests pass. Pinned it with `.nvmrc` and the `engines` field so nobody else hits the same wall. Left the `install-scripts approve` step out of the README since it was only a symptom of being on Node 26.

Reflection

Bryce: This was a good lesson in native modules being tied to a specific platform and Node version — the earlier attempt that built a Linux binary on a macOS project made that really clear. I now

understand why pinning the runtime matters on a shared repo instead of just letting everyone run whatever brew gave them.